Elia Trevisan, PhD

Robot Control @ Neura Robotics
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Motion planning and control engineer with expertise in gradient-free optimization methods (MPPI, CEM) now popular for model-based RL and world models. Background in system identification, control theory, autonomous driving, and manipulation. I love solving hard problems, from theory to hardware.

Experience

•Software Engineer - Robot Control *November 2026 - Present*

Neura Robotics, Stuttgart, Germany

- Developing motion planning and control software for humanoids, collaborative industrial arms, mobile manipulators, and autonomous mobile robots.

•PhD and Postdoc - Robot Motion Planning *October 2020 - June 2025*

TU Delft, Netherlands

- Contributed to optimization and sampling-based Model Predictive Control (MPC) (e.g., MPPI), Risk-Awareness, Machine Learning (ML) for trajectory prediction, and collaborated on Task and Motion Planning (TAMP).
- Applied planners to **vessels** (Roboat) in outdoor environments, **ground robots** (Jackal), and **mobile manipulators** (Boxer with Panda arm) for pushing and grasping using **Isaac Gym** as a dynamic model.
- **Mentored ten students**, of which six master thesis, one internship, one master research assignment, and two bachelor projects. All students were very successful, and **all** theses were **included in a publication**.
- Published several papers in **top robotics workshops, conferences and journals** (ICRA, IROS, MRS, RAL), with **two** nominated **Best Paper Finalists**.
- PhD **thesis** was a **finalist** for the **IEEE RAS TC-MRS Best Dissertation Award**.

•Teaching Assistant - Systems & Control *Sep 2019 - Jun 2020*

TU Delft, Netherlands

- Mentored master students in core control courses (Introduction Project, Nonlinear Systems Theory, Integration Projects), trusted by faculty to guide students through complex assignments and projects.

•Board Member - Systems & Control Student Association *Mar 2019 - Mar 2020*

DSA Kalman, TU Delft

- Organized course evaluations, maintained association website, and coordinated social and industry events.

•PLC Programmer - Industrial Automation *Mar 2018 - Aug 2018*

Tecnoquadri, Italy

- Designed PLC software for production lines and coordinated installation across factories. Hands-on experience with industrial robots and automation, wiring electrical panels, and mechanical systems.

Education

•MSc in Systems and Control *Sep 2018 - Aug 2020*

TU Delft, Netherlands ***Cum Laude***

- Courses on control theory, robust and multivariable control design, filtering & identification, optimization in systems and control, nonlinear systems theory, model predictive control, adaptive control, fault diagnosis and fault tolerant control, digital control.
- Graduated **top of my class**, thesis with Prof. Michel Verhaegen on modeling and identification for High Numerical-Aperture microscopy. Developed an algorithm that could both retrieve a clear image of the specimen and estimate the aberration in the microscope from very few out-of-focus images via alternating projections.

•BSc in Automation Engineering *Sep 2014 - Feb 2018*

University of Bologna, Italy and Tongji University, China

- **Won** AlmaTong, **a merit-based scholarship** to perform half of the bachelor's at Tongji University in Shanghai.
- Courses on automatic controls, mechanics, electronics, computer architectures, automatic machines, and robotics.
- Led a collaboration between my university and Electrolux and developed a simulator in Gazebo, using ROS, MoveIt, and Aruco markers to detect and manipulate household appliances with a UR5 Robot.

•Electrical Engineering High School *Sep 2009 - Jul 2014*

ITST J.F. Kennedy, Italy

- Theory and laboratories on electric machines, control, and design of civilian and industrial power systems.

Skills and Interests

Programming: Proficient in C/C++, Python (with PyTorch), Matlab, Simulink, Siemens TIA Portal, beginner in CUDA, HTML and CSS.

Robotics: Competent in ROS1 and ROS2, motion capture (Vicon and OptiTrack), experience with Lidar Odometry (KISS-ICP and LIO-SAM), Gazebo, Pybullet, IsaacGym, Docker, and several robotics platforms.

Development: Git, CI/CD pipelines, cloud storage (AWS S3), Linux.

Languages: Fluent in Italian and English, beginner in Spanish and Polish.

Personal InterestsI build stuff also outside working hours, but it's usually made out of wood. I competed in athletics many years ago. I now still enjoy running, but I recently got into bouldering. I love to hike, and I miss my Alps! I always snorkel when I get the chance. I've recently scuba-dived on the reef in Okinawa and have wanted to get a license since then. I used to play the drums as a teenager, and I am still always up for some good live music.